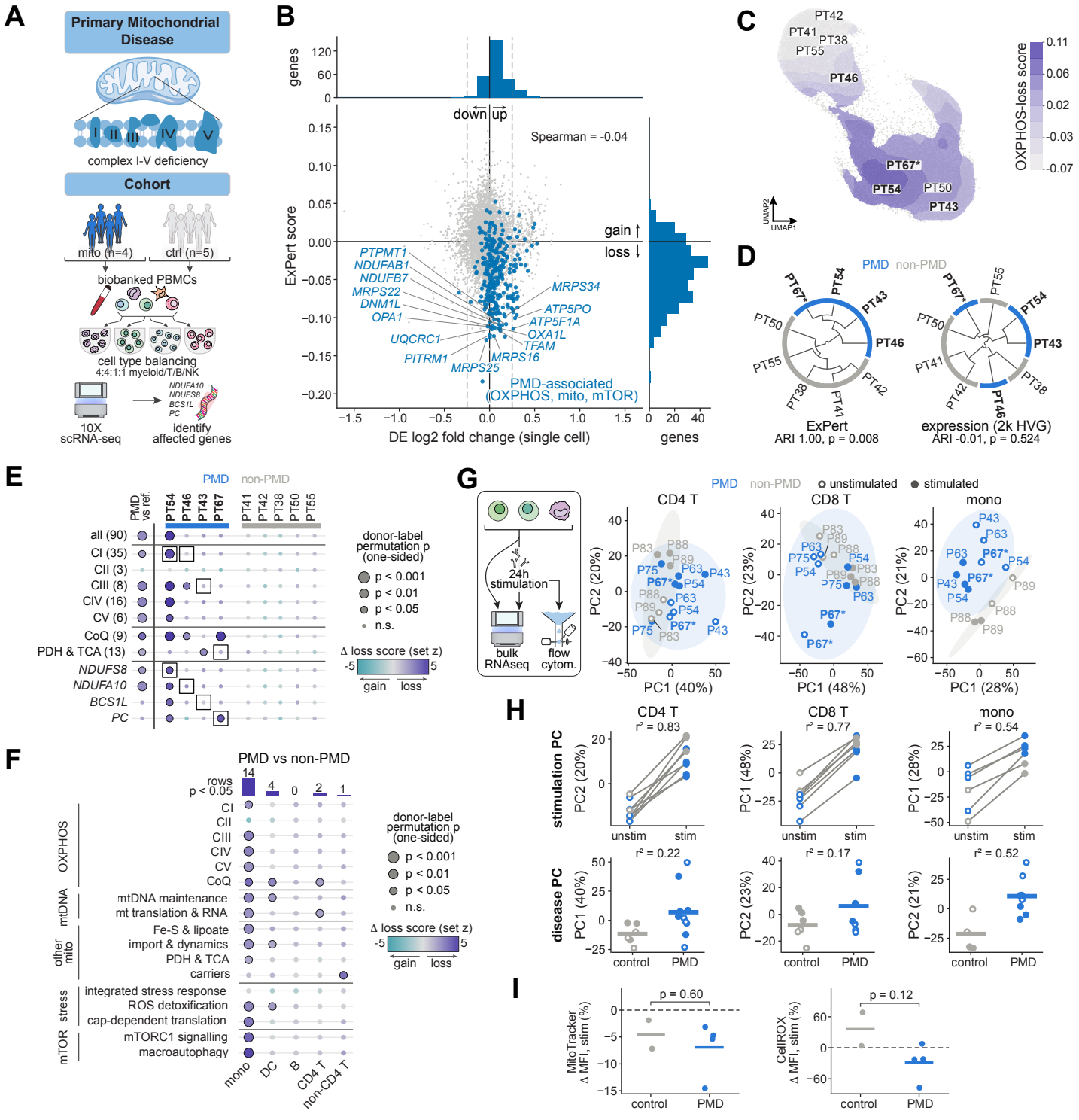
Figure 6


A
Primary Mitochondrial Disease
complex I-V deficiency
Cohort
mito (n=4)
ctrl (n=5)
biobanked PBMCs
cell type balancing
4:4:1:1 myeloid/T/B/NK
10X scRNA-seq
identify affected genes
B
genes
down
up
Spearman = -0.04
ExPert score
gain ↑
loss ↓
PMD-associated (OXPHOS, mito, mTOR)
DE log2 fold change (single cell)
C
OXPHOS-loss score
D
PMD non-PMD
ExPert
ARI 1.00, p = 0.008
expression (2k HVG)
ARI -0.01, p = 0.524
E
PMD non-PMD
donor-label permutation p (one-sided)
p < 0.001
p < 0.01
p < 0.05
n.s.
Δ loss score (set z)
gain
loss
F
PMD vs non-PMD
rows p < 0.05
OXPHOS
mtDNA
other mito
stress
mTOR
mtDNA maintenance
mt translation & RNA
Fe-S & lipoate
import & dynamics
PDH & TCA carriers
integrated stress response
ROS detoxification
cap-dependent translation
mTORC1 signalling
macroautophagy
mono
DC
B
CD4 T
non-CD4 T
G
24h stimulation
bulk RNAseq
flow cytom.
PMD non-PMD unstimulated stimulated
CD4 T
CD8 T
mono
PC1 (40%)
PC1 (48%)
PC1 (28%)
PC2 (20%)
PC2 (23%)
PC2 (21%)
H
stimulation PC
disease PC
unstim
stim
control
PMD
I
MitoTracker Δ MFI, stim (%)
CellROX Δ MFI, stim (%)
p = 0.60
p = 0.12